

Slide 1: Purpose of Normalization

23IT201 • Database Management Systems • Module 7: Normalization (1NF to BCNF)

Normalizing relations eliminates data redundancy and update anomalies while preserving functional dependencies and lossless joins.

Slide 2: First & Second Normal Forms

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1NF requires atomic attribute values. 2NF eliminates partial dependencies (non-prime attributes must depend on the full primary key).

Slide 3: 3NF vs BCNF

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3NF eliminates transitive dependencies. BCNF requires that for every non-trivial FD $X \rightarrow Y$, X must be a superkey.